

2025 DYNAMICS CON

A vibrant, stylized illustration of a city skyline at sunset. The sky transitions from a deep purple at the top to a bright yellow-orange near the horizon. Silhouettes of various skyscrapers in shades of blue and purple are scattered across the background. In the foreground, a large, grey, curved structure resembling a stylized wave or a piece of modern architecture sits on a dark purple base. To the right, a fountain with multiple jets of water is depicted. Two superheroes are seated on the fountain's base: one in a blue and red suit on the left, and another in a purple and blue suit on the right. The overall aesthetic is bold and comic-book inspired.

dynamicscon.com

Business Central Performance Management



**2025
DYNAMICS
CON**



Marcel Chabot

Development Practice Manager

A life-long software developer and “maker”, Marcel brings experience and a unique approach to complex problems and solutions in D365



Session Goals

- Identify tools and technology to analyze Business Central Performance
- Understand the limitations of each tool set
- Breakdown the common barriers to good Business Central Performance
- Have a good time and learn something



Session Agenda

- **Tools and Technology**
 - In Business Central Performance Profiler
 - Application Insights
 - Visual Studio Code
- **Reporting and Analysis**
 - KQL Reporting
 - Power BI Dashboards
- **Case Study Analysis**
- **Q&A**





Tool 1
Performance
Profiler

Business Central Performance Profiler

- Built-in to Business Central
 - Also called the In-Client Performance Profiler
- Useful to isolate a repeatable problem
- Can be used to benchmark a process
- Limited to 10 minutes or 100mb whichever comes first.



CRONUS USA, Inc.

Customers Vendors

✕ This is a sandbox environment

Get started

Hi, meet B

You're all set to try
Cronus. Go explore
quick tour first.

Tell me what you want to do



Performance Profiler

Go to Pages and Tasks

> Performance Profiler

Administration

Search for 'Performance Profiler'

📄 Search company data

🔗 Search Help

Didn't find what you were looking for? Try [exploring pages](#) or [exploring reports](#)





✓ Saved



Performance Profiler

▶ Start ■ Stop ✕ Clear ↓ Download 📄 Upload ⚙ Settings

Troubleshoot slow operations

Use the Start and Stop buttons to record a process that you think is slow. The charts will show how much time each app used.

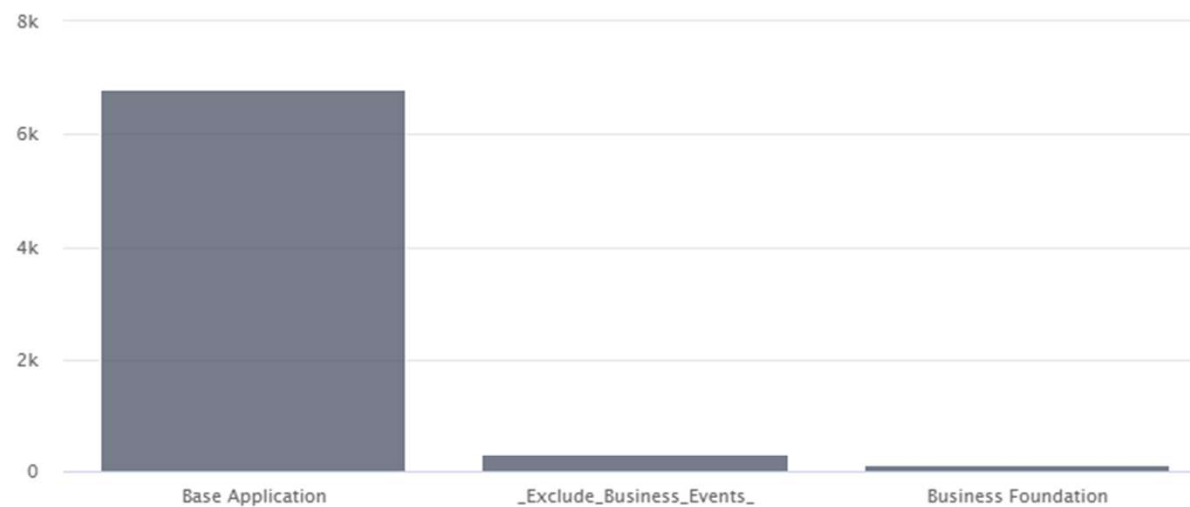


Performance Profiler

▶ Start ■ Stop ✕ Clear ⬇ Download 📁 Upload ⚙ Settings

Active Apps

Actions



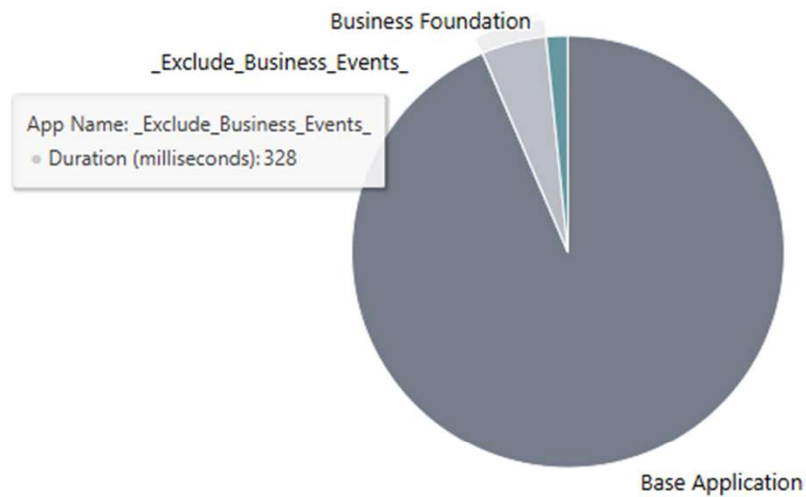
Shows the apps that were active during the recording. They were either running or called other apps. The duration represents the time you might save by removing an app.



Show technical information ☒

Time Spent

Actions



Shows which apps were running during the recording. Durations are self-time. They show the length of activity but do not include time spent calling other apps.



Time Spent by Application Object

Object Type	Object Name	Time Spent ↓	App Name
→ Codeunit	⋮ Sales Invoice Aggregator	2 seconds 101 milliseconds	Base Application
Table	Sales Line	438 milliseconds	Base Application
Codeunit	Sales-Post	427 milliseconds	Base Application
Codeunit	External Events Helper	328 milliseconds	_Exclude_Business_Events_
Codeunit	Sales Tax Calculate	328 milliseconds	Base Application
Codeunit	Gen. Jnl.-Post Line	327 milliseconds	Base Application
Codeunit	Inventory Adjustment	219 milliseconds	Base Application
Table	Sales Header	218 milliseconds	Base Application
Table	General Posting Setup	113 milliseconds	Base Application
Codeunit	Find Record Management	112 milliseconds	Base Application
Codeunit	Sales - Calc Discount By Type	112 milliseconds	Base Application



Call Tree

Method Name	Object Type	Object Name	Self Time	Total Time	App Name
Post - OnAction	Page	Sales Invoice		4 seconds 602 mi...	Base Application
PostDocument	Page	Sales Invoice		4 seconds 602 mi...	Base Application
SendToPosting	Table	Sales Header		4 seconds 602 mi...	Base Application
OnRun	Codeunit	Sales-Post (Yes/No)		4 seconds 602 mi...	Base Application
Code	Codeunit	Sales-Post (Yes/No)		4 seconds 602 mi...	Base Application
ConfirmPost	Codeunit	Sales-Post (Yes/No)	110 milliseconds	110 milliseconds	Base Application
RunSalesPost	Codeunit	Sales-Post (Yes/No)		4 seconds 492 mi...	Base Application
OnRun	Codeunit	Sales-Post		4 seconds 492 mi...	Base Application
RunWithCheck	Codeunit	Sales-Post	112 milliseconds	4 seconds 492 mi...	Base Application
CheckAndUpdate	Codeunit	Sales-Post		2 seconds 741 mi...	Base Application
CheckSalesDocument	Codeunit	Sales-Post	100 milliseconds	218 milliseconds	Base Application





Choose the sampling interval of the performance profiler. Smaller intervals will result in more precise timings, but it might result in higher load of the system.

- ☐ 50 ms
- ☒ 100 ms
- ☐ 150 ms

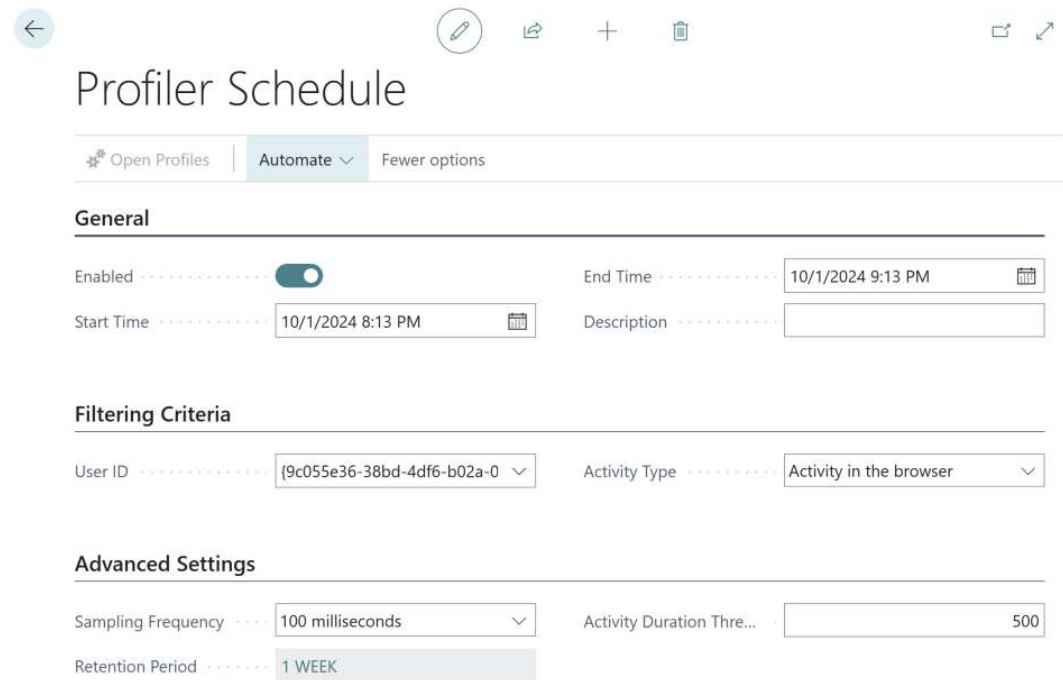
OK

Cancel



Sneaky Issues

- Schedule to capture performance issues
 - Time
 - User
 - Duration



The screenshot shows the 'Profiler Schedule' configuration window. At the top, there are navigation icons: a back arrow, an edit icon, a share icon, a plus icon, a trash icon, and a refresh icon. Below the title bar, there are three tabs: 'Open Profiles' (with a star icon), 'Automate' (selected), and 'Fewer options'. The 'General' section contains an 'Enabled' toggle switch (turned on), an 'End Time' field set to '10/1/2024 9:13 PM' with a calendar icon, a 'Start Time' field set to '10/1/2024 8:13 PM' with a calendar icon, and a 'Description' text input field. The 'Filtering Criteria' section has a 'User ID' dropdown menu showing '[9c055e36-38bd-4df6-b02a-0]' and an 'Activity Type' dropdown menu showing 'Activity in the browser'. The 'Advanced Settings' section includes a 'Sampling Frequency' dropdown set to '100 milliseconds', an 'Activity Duration Thre...' field set to '500', and a 'Retention Period' dropdown set to '1 WEEK'.

Profiler Schedule

Open Profiles | Automate | Fewer options

General

Enabled ☒ End Time 10/1/2024 9:13 PM

Start Time 10/1/2024 8:13 PM Description

Filtering Criteria

User ID [9c055e36-38bd-4df6-b02a-0] Activity Type Activity in the browser

Advanced Settings

Sampling Frequency 100 milliseconds Activity Duration Thre... 500

Retention Period 1 WEEK





Tool 2
VS Code Snapshot

Visual Studio Code

- Attaches to a running Business Central
- Can replay the issue over and over again
- Deep and detailed analysis
- 10 Minute or 100mb limit
- Can have a large performance impact!





✓ Saved



User Card

Effective Permissions

Send Email

Sent Emails

More options



General

Show more

User Name ADMIN

Full Name

Status Active

Contact Email

Microsoft 365

Authentication Email

Authentication Status Disabled

Web Service

Web Service Access Key



User Permission Sets

New Line

Delete Line

Permissions



Permission Set ↑	Description	Company ↑	Extension Name	Permission Scope
SUPER	This role has all per...			System
→ D365 SNAPSHOT DEBUG	Snapshot Debug		System Application	System





Information

Help & Support

The demonstration and setup data in CRONUS evaluation companies and My Company, including the predefined configuration packages, are not designed to satisfy local requirements in any country or region.

Give feedback

Do you have a great idea that you would like to see in Business Central?

Register your idea and get others in the community to vote for it at aka.ms/BusinessCentralIdeas

Troubleshooting

Version: US Business Central 25.0 (Platform 25.0.24850.0 + Application 25.0.23364.24925)

[View the last known error](#)

[Inspect pages and data](#)

[Analyze performance](#)

[Analyze performance with scheduled profiler](#)

[Enable additional logging](#)

[Attach debugger to this session](#)

Report a problem

[Copy the text below](#) and add it to your support request:

Microsoft Entra tenant ID: (blank), Environment: Sandbox (Sandbox)

Session ID (client): 378f883c-b473-4ef1-aa00-9cee9b4d456f

Session ID (server): 158

User telemetry ID: 88049e18-72a8-45d7-9c13-2e6fab7663d7

Help

About this task or page

Learn more about your current task or page Information

[Go to Microsoft Learn](#)

Related resources from Microsoft Learn

[Resources for help and support](#)

Other resources

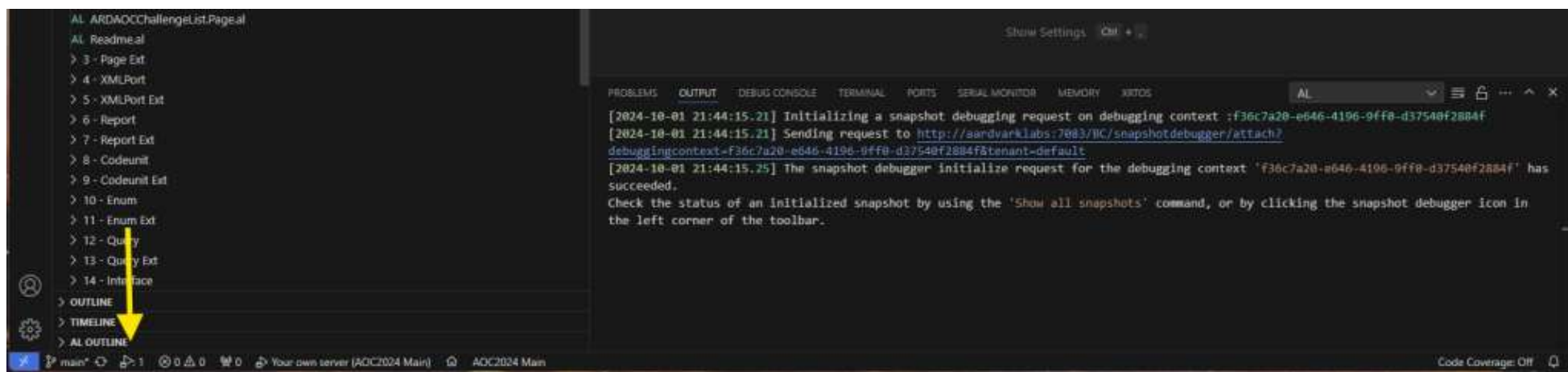
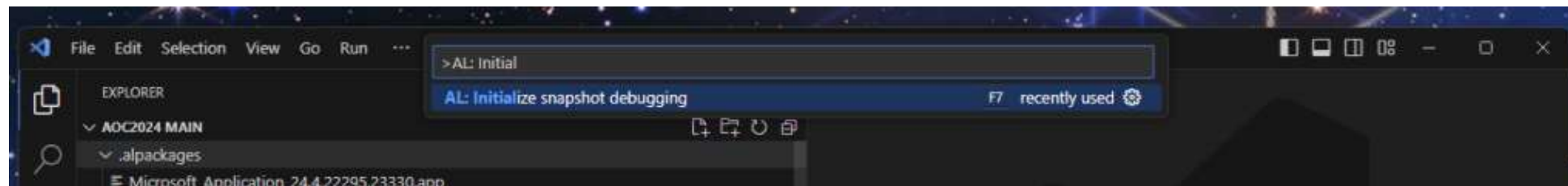
[Help & Support](#)

[Keyboard Shortcuts](#)

[Community](#)

```
1 {
2   "version": "0.2.0",
3   "configurations": [
4     {
5       "name": "snapshotInitialize: Your own server",
6       "type": "al",
7       "request": "snapshotInitialize",
8       "sessionId": 158,
9       "environmentType": "OnPrem",
10      "server": "http://aardvarklabs",
11      "tenant": "default",
12      "serverInstance": "BC",
13      "authentication": "UserPassword",
14      "breakOnNext": "WebClient",
15      "executionContext": "DebugAndProfile",
16      "profilingType": "Instrumentation"
17    }
18  ]
19 }
```







Customer Card



✓ Saved



10000 · Adatum Corporation

[Home](#)[Request Approval](#)[New Document](#)[Prices & Discounts](#)[Customer](#)[Report](#)[More options](#)

Contact

Apply Template

Merge With...

Send Email



General >

Adatum Corporation

Address & Contact

[Show less](#)

Address

Address

Address 2

Country/Region Code

City

State

ZIP Code

[Show on Map](#)

Mobile Phone No

es@contoso.com



You cannot send the email.
Verify that the email settings are correct.

[Share details](#) ▾

Was this information helpful? Yes No

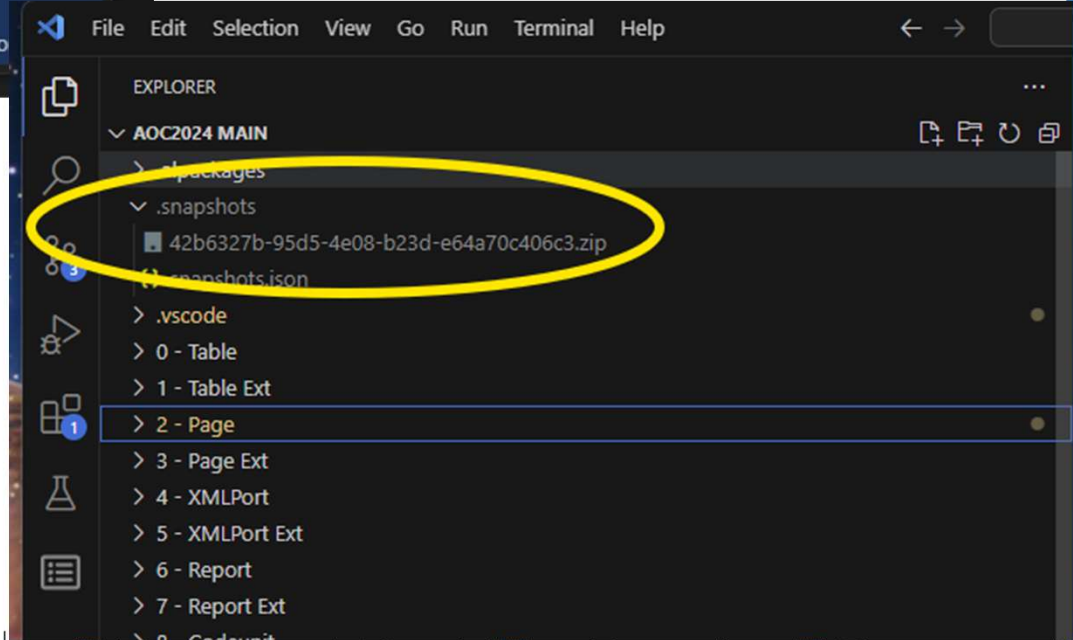
[OK](#)

Contact

Contact Code

```
>AL: Finish|
AL: Finish snapshot debugging on the server Alt + F7 recently used
vscode > launch icon > Launch Targets > snapshotInitialize: Your own server
```

```
The snapshot debug sessions
42b6327b-95d5-4e08-b23d-e64a70c406c3
status:Started tenant:default user:The first user with a session on the tenant kind:UserSession
1 {
```



File Edit Selection View Go Run ...

EXPLORER

- ▼ AOC2024 MAIN
 - > .alpackages
 - ▼ .snapshots
 - 42b6327b-95d5-4e08-b23d-e64a70c406c3.zip
 - ! snapshots.json
 - ▼ .vscode
 - > .alcache
 - ! launch.json
 - ! rad.json
 - > 0 - Table
 - > 1 - Table Ext
 - > 2 - Page
 - > 3 - Page Ext
 - > 4 - XMLPort
 - > 5 - XMLPort Ext
 - > 6 - Report
 - > 7 - Report Ext
 - > 8 - Codeunit
 - > 9 - Codeunit Ext
 - > 10 - Enum
 - > 11 - Enum Ext
 - > 12 - Query
 - > 13 - Query Ext
 - > 14 - Interface
 - > 15 - ControlAddin
 - > 16 - Profiles
 - > 17 - Permission Set
 - > 18 - Permission Set Extension
 - > Layouts
 - > Translations
 - Aardvark Labs Aardvark Labs Advent of Co... M
 - Aardvark Labsicon.jpg
 - ! app.json
 - > OUTLINE
 - > TIMELINE
 - > AL OUTLINE

The snapshot debug sessions

42b6327b-95d5-4e08-b23d-e64a70c406c3

status:Downloaded created:Oct 1, 2024, 09:47:02 PM

The file is not displayed in the text editor because it is either binary or uses an unsupported text encoding.

Open Anyway

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SERIAL MONITOR MEMORY XRTOS

[2024-10-01 21:51:09.31] Finishing a snapshot debugger attach request on debugging context : '42b6327b-95d5-4e08-b23d-e64a70c406c3'

[2024-10-01 21:51:09.31] Sending request to <http://aardvarklabs:7083/BC/snapshotdebugger/finish?debuggingcontext=42b6327b-95d5-4e08-b23d-e64a70c406c3&tenant=default>

[2024-10-01 21:51:09.57] The snapshot debugger attach session has finished. Snapshot successfully saved as 'd:\AardvarkLabs\AOC2024\AOC2024 Main\snapshots\42b6327b-95d5-4e08-b23d-e64a70c406c3.zip'.

As the snapshot can contain customer privacy data, you should handle it according to privacy compliance and remember to delete it when you are done using it.

Use the 'Show all snapshots' command, or by clicking the snapshot debugger icon in the left corner of the toolbar to view the downloaded snapshots ready for debugging.

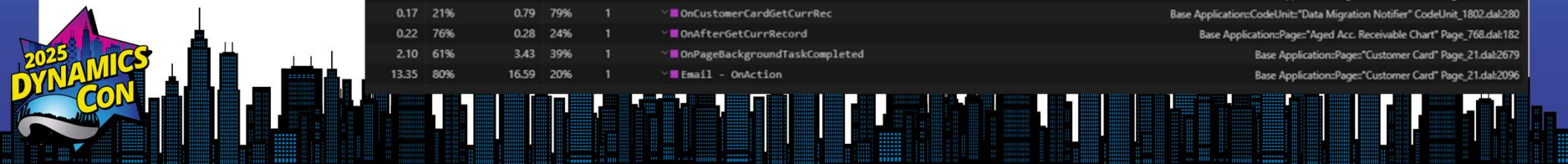
main 816.80KB Code Coverage: Off





- AL: Generate Profile File

Self Time in ms	Total Time in ms	Hit count	Function Name	
3.23	89%	3.62	11%	2
25.76	86%	29.70	14%	12
1.01	20%	4.96	80%	2
2.64	100%	2.64	0%	8
6.46	93%	6.89	7%	6
0.12	100%	0.12	0%	2
9.75	44%	21.71	56%	2
47.91	9%	512.49	91%	1
22.76	72%	31.33	28%	1
2.74	30%	8.99	70%	1
16.04	100%	16.04	0%	1
2.09	13%	15.48	87%	1
4.05	18%	21.98	82%	1
0.71	51%	1.38	49%	1
15.66	100%	15.66	0%	1
11.49	97%	11.77	3%	1
3.08	47%	6.46	53%	1
0.64	68%	0.94	32%	1
0.00	0%	3.03	100%	1
18.73	50%	37.22	50%	1
3.80	90%	4.18	10%	1
3.99	30%	12.94	70%	1
0.20	60%	0.33	40%	1
0.24	100%	0.24	0%	3
0.31	100%	0.31	0%	3
5.67	95%	5.97	5%	3
1.87	73%	2.53	27%	1
0.17	21%	0.79	79%	1
0.22	76%	0.28	24%	1
2.10	61%	3.43	39%	1
13.35	80%	16.59	20%	1
				Base Application:Pages:"O365 Activities" Page_1310.dal:376
				Base Application:Tables:"Activities Cue" Table_1313.dal:223
				Base Application:Pages:"O365 Activities" Page_1310.dal:370
				System Application:Pages:"Checklist Banner" Page_1990.dal:368
				System Application:CodeUnit:"Checklist Banner Impl." CodeUnit_1996.dal:78
				System Application:Pages:"Checklist Banner" Page_1990.dal:362
				Aardvark Labs Advent of Code 2024:Page:ARD_AOCChallengeCard ARDAOCChallengeCard.Page.al:178
				Aardvark Labs Advent of Code 2024:Page:ARD_AOCChallengeCard ARDAOCChallengeCard.Page.al:149
				Base Application:Pages:"Customer List" Page_22.dal:1713
				Base Application:Pages:"Customer List" Page_22.dal:1729
				Dynamics GP Historical Data:PageExtension:"Customer List Ext." PageExtension_41018.dal:71
				Base Application:CodeUnit:"Data Migration Notifier" CodeUnit_1802.dal:35
				Base Application:Pages:"Customer List" Page_22.dal:1696
				Base Application:CodeUnit:"Data Migration Notifier" CodeUnit_1802.dal:274
				Dynamics GP Historical Data:PageExtension:"Hist. Sales Factbox Ext." PageExtension_41021.dal:46
				Base Application:Pages:"Customer Statistics FactBox" Page_9082.dal:220
				Base Application:Pages:"Document Attachment Factbox" Page_1174.dal:74
				Base Application:Pages:"Customer Statistics FactBox" Page_9082.dal:281
				Base Application:Pages:"Doc. Attachment List Factbox" Page_1178.dal:250
				Base Application:Pages:"Doc. Attachment List Factbox" Page_1178.dal:260
				Base Application:Pages:"Customer Card" Page_21.dal:2547
				Base Application:Pages:"Customer Card" Page_21.dal:2573
				Base Application:Pages:"Aged Acc. Receivable Chart" Page_768.dal:197
				Base Application:Pages:"Customer Card" Page_21.dal:2873
				Base Application:Tables:Customer Table_18.dal:2593
				Base Application:CodeUnit:"Customer Mgt." CodeUnit_1302.dal:315
				Base Application:Pages:"Customer Card" Page_21.dal:2498
				Base Application:CodeUnit:"Data Migration Notifier" CodeUnit_1802.dal:280
				Base Application:Pages:"Aged Acc. Receivable Chart" Page_768.dal:182
				Base Application:Pages:"Customer Card" Page_21.dal:2679
				Base Application:Pages:"Customer Card" Page_21.dal:2096



@t > 500

Self Time in ms	Total Time in ms	Hit count	Function Name	
47.91	9%	512.49	91%	1
464.59	91%	510.41	9%	1

CalculateResult1 - OnAction
 CalculateResult1

Aardvark Labs Advent of Code 2024:Page:ARD_AOCChallengeCard ARDAOCChallengeCard.Page.al:149
 Aardvark Labs Advent of Code 2024:CodeUnit:ARD_ResultsCalculator ARDResultsCalculator.CodeUnit.al:7

@da ~ = Aardv*

Self Time in ms	Total Time in ms	Hit count	Function Name	
9.75	44%	21.71	56%	2
6.61	70%	9.35	30%	2
0.20	32%	0.62	68%	2
0.22	34%	0.63	66%	2
4.92	87%	5.62	13%	2
47.91	9%	512.49	91%	1
464.59	91%	510.41	9%	1

OnAfterGetCurrRecord
 RetrieveChallenge1Text
 RetrieveChallenge2Text
 RetrieveChallengeExample
 RetrieveChallengeData
 CalculateResult1 - OnAction
 CalculateResult1

Aardvark Labs Advent of Code 2024:Page:ARD_AOCChallengeCard ARDAOCChallengeCard.Page.al:178
 Aardvark Labs Advent of Code 2024:Table:ARD_AOCChallenge ARDAOCChallenge.Table.al:79
 Aardvark Labs Advent of Code 2024:Table:ARD_AOCChallenge ARDAOCChallenge.Table.al:96
 Aardvark Labs Advent of Code 2024:Table:ARD_AOCChallenge ARDAOCChallenge.Table.al:113
 Aardvark Labs Advent of Code 2024:Table:ARD_AOCChallenge ARDAOCChallenge.Table.al:130
 Aardvark Labs Advent of Code 2024:Page:ARD_AOCChallengeCard ARDAOCChallengeCard.Page.al:149
 Aardvark Labs Advent of Code 2024:CodeUnit:ARD_ResultsCalculator ARDResultsCalculator.CodeUnit.al:7



An illustration featuring a blue city skyline with various skyscrapers. In the foreground, a white sign with a black border is mounted on a grey pole. The sign contains the text 'Tool 3 Application Insights'. The background is a blue sky with horizontal stripes and a silhouette of a person swimming in the upper right corner.

Tool 3
Application
Insights

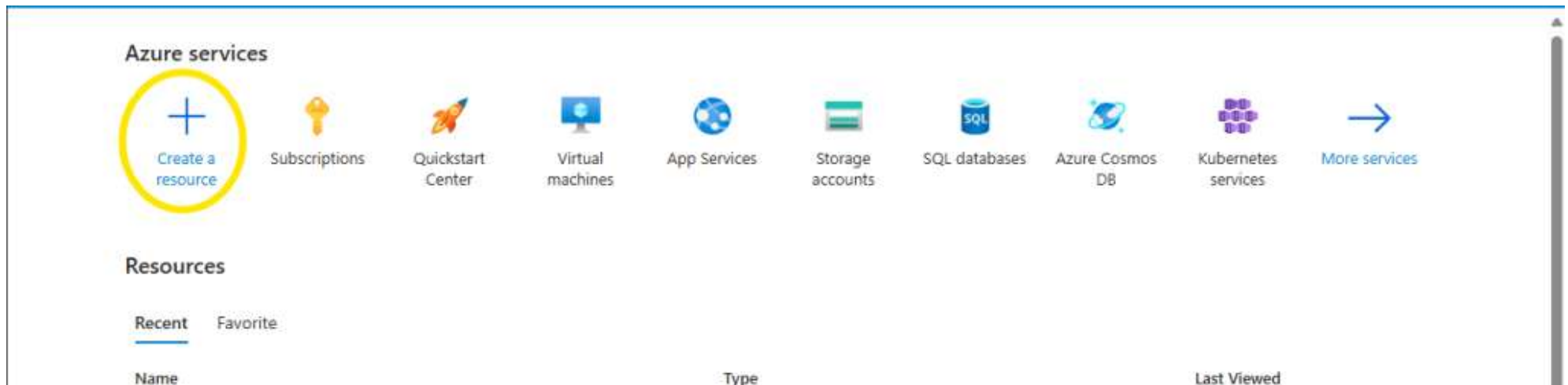
Application Insights

- Azure Service
- Long term performance tracking
- Defaults to monitoring 30 days of data
- Can cost money, typically less than a Starbucks run.



Step 1: Application Insights

- Portal.Azure.com



[Home](#) >

Create a resource ...



Get Started

Recently created

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application insights

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Essentials 50K



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Private plans

Categories

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IT & Management Tools
(165)

AI + Machine Learning (87)



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Pricing : All

Operating System : All

Publisher Type : All

Product Type : All

☐ Azure services only

Publisher name : All

Showing 1 to 20 of 611 results for 'Application Insights'. [Clear search](#)

Tile view



Application Insights

Microsoft

Azure Service

Application performance, availability and usage information at your fingertips.

Create



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★ 4.3 (525 ratings)

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Application performance, availability and usage information at your fingertips

Is my website available at all times?

Is the user experience responsive and usable?

How are customers using my app/website?

Application Insights answers these questions and many more. With Application Insights, you have full observability into your application across all components and dependencies of your complex distributed architecture, as well as:

- Availability and performance monitoring
- Deep code diagnostics with profiler and debugger
- Users and usage insights
- Out of the box performance and failures triage with end-to-end transaction diagnostics
- Seamless integration with Microsoft Azure and Visual Studio
- Supports ASP.NET, .NET Core, Java, Node.js applications and websites on Azure or on-premises



Application Insights ...

Monitor web app performance and usage

Basics Tags Review + create

Create an Application Insights resource to monitor your live web application. With Application Insights, you have full observability into your application across all components and dependencies of your complex distributed architecture. It includes powerful analytics tools to help you diagnose issues and to understand what users actually do with your app. It's designed to help you continuously improve performance and usability. It works for apps on a wide variety of platforms including .NET, Node.js and Java EE, hosted on-premises, hybrid, or any public cloud. [Learn More](#)

PROJECT DETAILS

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ	Azure subscription 1 ▼
Resource Group * ⓘ	(New) AppInsights ▼

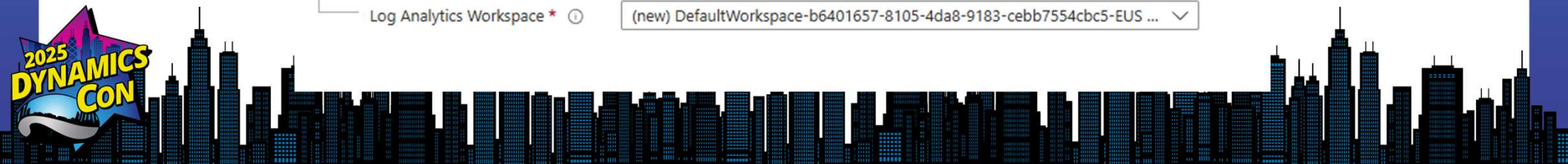
[Create new](#)

INSTANCE DETAILS

Name * ⓘ	BCMonitoring ✓
Region * ⓘ	(US) East US ▼

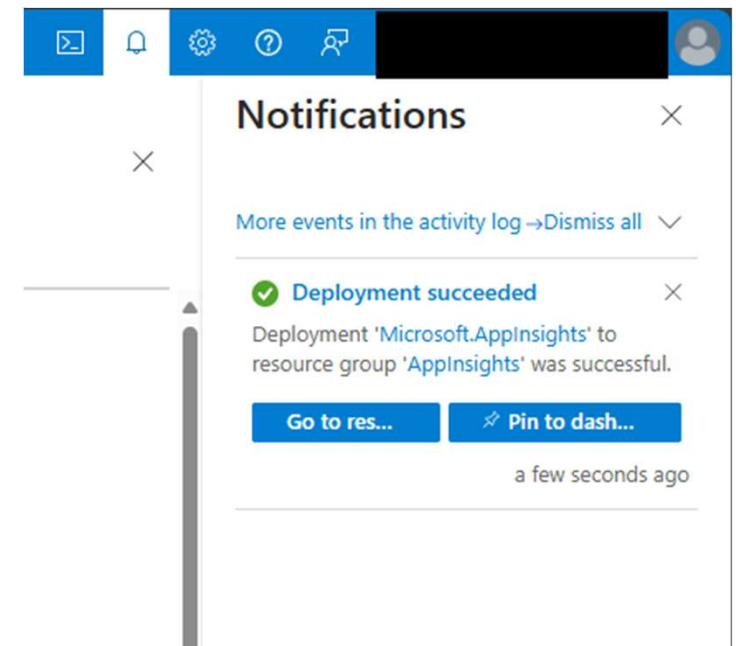
WORKSPACE DETAILS

Subscription * ⓘ	Azure subscription 1 ▼
Log Analytics Workspace * ⓘ	(new) DefaultWorkspace-b6401657-8105-4da8-9183-cebb7554cbc5-EUS ... ▼



Finalize Application Insights

- Review + Create



Home >



BCMonitoring

Application Insights



Search

★ Application Dashboard Getting started Search Logs Monitor resource group Feedback Favorites Rename ...

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

> Investigate

> Monitoring

> Usage

> Configure

> Settings

> Automation

> Help

Essentials

[JSON View](#)

Resource group [\(move\)](#)

[AppInsights](#)

Location

East US

Subscription [\(move\)](#)

Subscription ID

[Tags \(edit\)](#)

[Add tags](#)

Instrumentation Key

Connection String

InstrumentationKey= ingestionE...

Workspace

[DefaultWorkspace](#) -EUS

Show data for last:

30 minutes **1 hour** 6 hours 12 hours 1 day 3 days 7 days 30 days

Failed requests



100

80

Server response time



100ms

80ms



Back to Business Central

Dynamics 365 Business Central admin center

Environments > [Redacted]

Details Active ⓘ

Name [Redacted]	Application Family Business Central	Country/region US	Azure Region South Central US
Type Production	Telemetry ⓘ Not Set (Define)	Security Group Not Set (Define)	Access with Microsoft 365 Licenses ⓘ Off (Modify)
URL [Redacted]			



Configure BC

- Paste in Connection String
- BC will reboot!!!

Logging telemetry

Enter the connection string for the Azure Application Insights resource where you want telemetry to be stored for the environment.

[Learn more](#)

Caution: Applying this connection string requires a restart to the environment. We recommend doing this when no users are active in Business Central.

Enable telemetry



Azure Application Insights Connection String *

InstrumentationKey=[REDACTED];IngestionEndp...

Save

Cancel



Home > All resources > BC_Production_App_Insights

BC_Production_App_Insights | Logs

Application Insights

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
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- Monitoring
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 - Diagnostic settings
 - Logs**
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- Automation
- Help

New Query 1

BC_Production_Ap... Select scope

Run

Time range : Last 24 hours

Save

Tables Queries Functions ...

Search

Filter Group by: Solution

Collapse all

Favorites

You can add favorites by clicking on the ☆ icon

Application Insights

- availabilityResults
- browserTimings
- customEvents
- customMetrics
- dependencies
- exceptions
- pageViews
- performanceCounters
- requests
- traces

1 Type your query here or click one of the q

Query history



Tracked Metrics

- Browser Timings
- Page Views
- Traces



The Kusto Query aka KQL

- The query language used to parse Telemetry files
- Useful to filter through the scads of data
- Very useful when exporting the query to Power BI



Run Time range : Set in query Show : Max. KQL mode							
<pre> 1 traces 2 where timestamp > ago(30d) 3 where customDimensions.alObjectName != '' 4 project 5 timestamp, 6 session_Id, 7 customDimensions.sessionId, 8 user_Id, 9 message, 10 customDimensions.companyName, customDimensions.clientType, customDimensions.alObjectName, customDimensions.alObjectId, 11 customDimensions.alObjectType, customDimensions.executionTime, customDimensions.totalTime, customDimensions.extensionVersion, 12 customDimensions.extensionId, customDimensions.extensionName, customDimensions.extensionPublisher, customDimensions.environmentName, 13 customDimensions.environmentType, customDimensions.aadTenantId, customDimensions.alStackTrace, customDimensions.sqlStatement, 14 customDimensions.failureReason, customDimensions.alErrorMessage 15 sort by timestamp desc 16 </pre>							
Results Chart							
timestamp [UTC]	session_Id	user_Id	message	customDimensions_compan...	customDimensions_clientTy...	customDimensions_alObjec...	customDimensions_alObj...
> 3/21/2025, 3:54:37.034 AM	680507ad-8319-487d-ada1-50...	N/A	Upgrade tag searched for: MS-...	CRONUS USA, Inc.	Background	Upgrade Tag Impl.	9996
> 3/21/2025, 3:54:37.052 AM	680507ad-8319-487d-ada1-50...	N/A	Upgrade tag searched for: MS-...	CRONUS USA, Inc.	Background	Upgrade Tag Impl.	9996
> 3/20/2025, 4:05:50.857 AM	a7d2f823-115c-423e-8647-b2d...	N/A	Upgrade tag searched for: MS-...	My Company	Background	Upgrade Tag Impl.	9996
> 3/20/2025, 4:05:50.271 AM	a7d2f823-115c-423e-8647-b2d...	N/A	Upgrade tag searched for: MS-...	CRONUS USA, Inc.	Background	Upgrade Tag Impl.	9996
> 3/20/2025, 4:00:12.734 AM	7f239aa4-3f20-4596-8163-b86...	37c6de19-fa1e-4b1b-ace2-0ba...	Job queue entry finished: B6DB...	CRONUS USA, Inc.	Background	System Telemetry Logger	8713
> 3/20/2025, 4:00:12.734 AM	7f239aa4-3f20-4596-8163-b86...	37c6de19-fa1e-4b1b-ace2-0ba...	Job queue entry enqueued: B6...	CRONUS USA, Inc.	Background	System Telemetry Logger	8713



Analyze and monitor telemetry with KQL

<https://learn.microsoft.com/en-us/dynamics365/business-central/dev-itpro/administration/telemetry-analyze-with-kql#how-can-i-query-telemetry-from-log-analytics>





Tool 4
Power BI
Dashboard

Power BI Reporting

The screenshot displays the Power BI web interface. At the top, there's a header with 'New Query 1*' and a '+'. Below it, a navigation bar includes 'BC_Production_Ap...', 'Select scope', a 'Run' button, 'Time range: Last 24 hours', 'Save', 'Share', 'New alert rule', 'Export', 'Pin to', and 'Format query'. The 'Export' menu is open, showing options: 'CSV (all columns)', 'CSV (displayed columns)', 'Open in Excel', 'Power BI (as an M query)', and 'Power BI (new Dataset)'. A yellow arrow points to the 'Power BI (as an M query)' option. On the left, there's a sidebar with 'Tables', 'Queries', and 'Functions' tabs, a search bar, and a 'Filter' button. The main area shows a table with one row labeled 'traces'.

Try the new Log Analytics

BC_Production_Ap... Select scope Run Time range : Last 24 hours Save Share New alert rule Export Pin to Format query

Tables Queries Functions ...

Search

Filter Group by: Solution

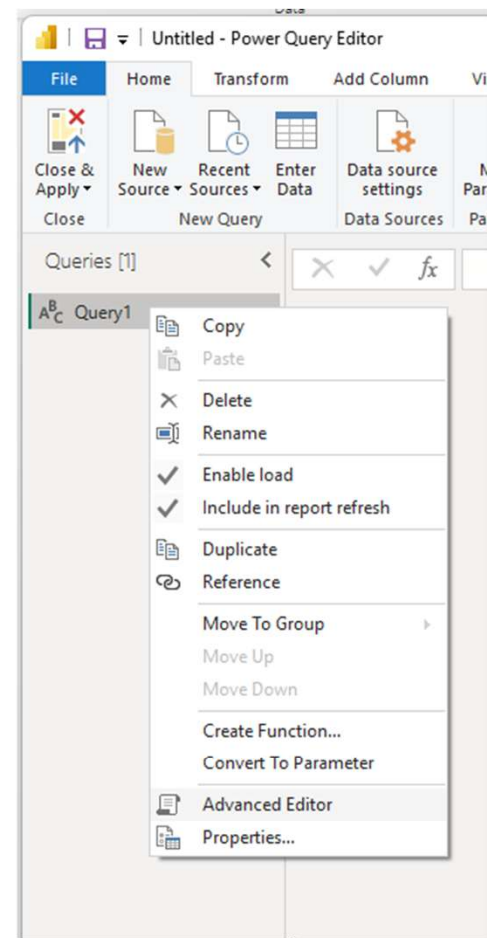
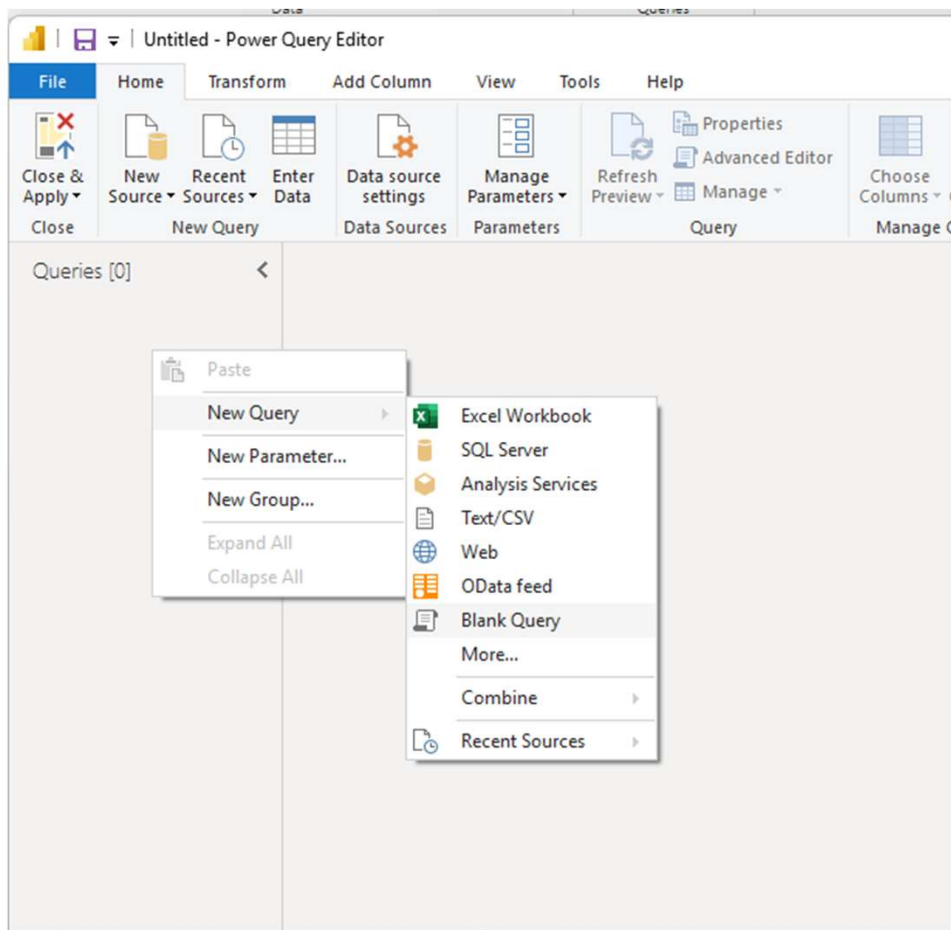
Collapse all

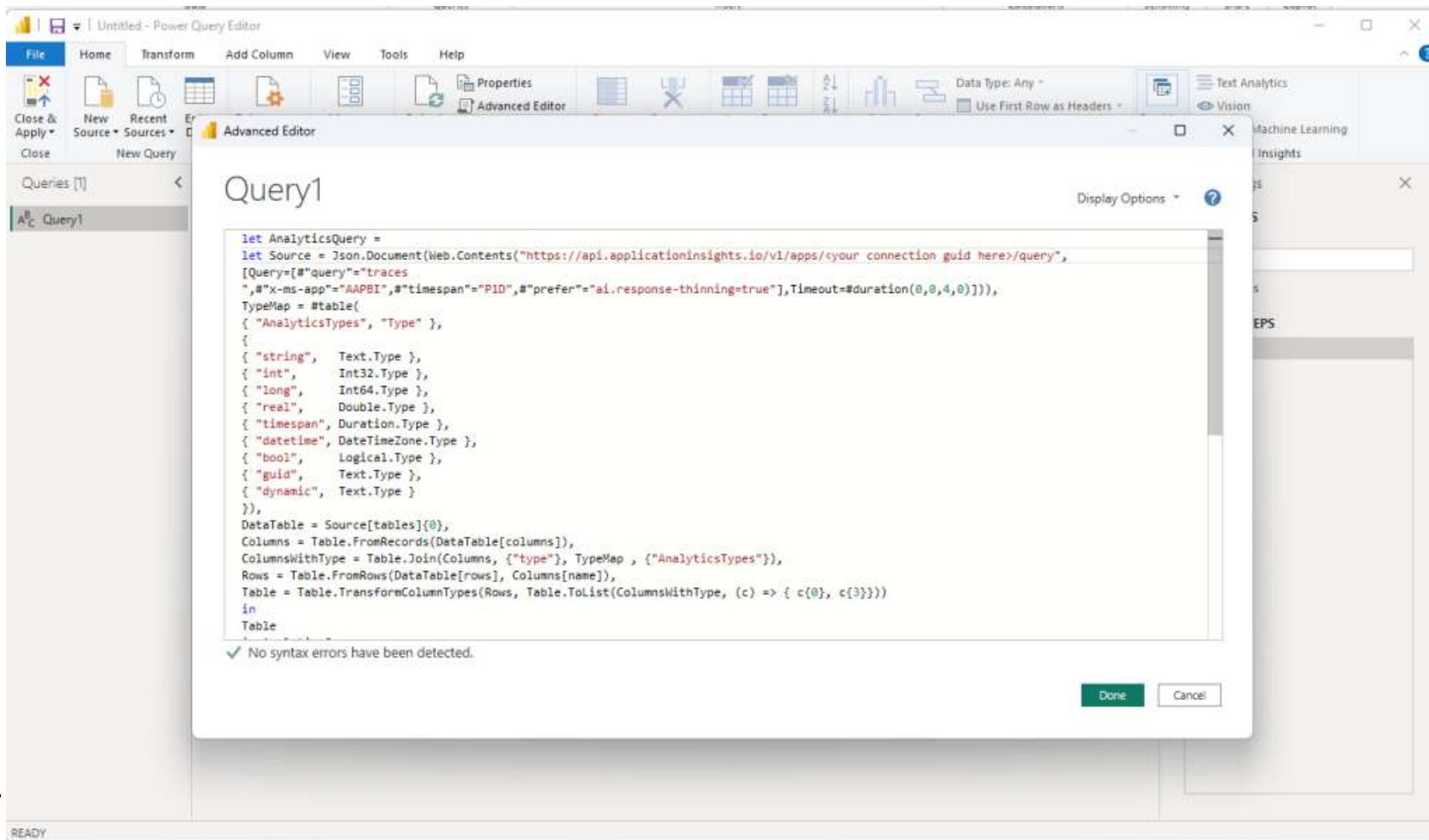
Favorites

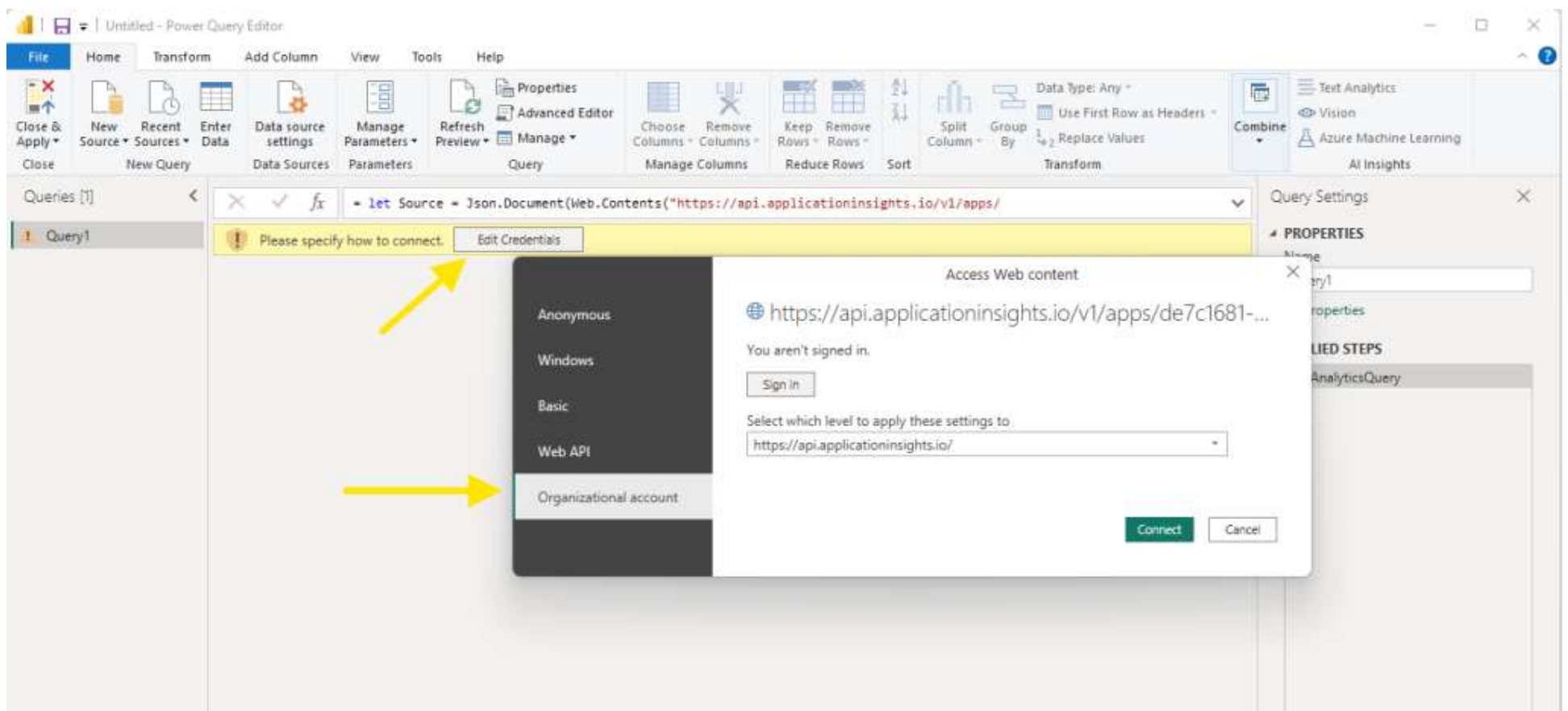
1 traces

- CSV (all columns)
- CSV (displayed columns)
- Open in Excel
- Power BI (as an M query)
- Power BI (new Dataset)









Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Advanced Editor Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Headers Replace Values Transform Combine Text Analytics Vision Azure Machine Learning AI Insights

Queries [1] X ✓ /x = Table.TransformColumns(AnalyticsQuery,{{"customDimensions", Json.Document}})

Traces	Item Type	customDimensions	customMeasurements	operation
1	trace	Record	null	Success re
2	trace	Record	null	Authoriza
3	trace	Record	null	Web Serv
4	trace	Record	null	Authoriza
5	trace	Record	null	Authoriza
6	trace	Record	null	Authoriza
7	trace	Record	null	Web Serv
8	trace	Record	null	Authoriza
9	trace	Record	null	Web Serv
10	trace	Record	null	Web Serv
11	trace	Record	null	Web Serv
12	trace	Record	null	Authoriza
13	trace	Record	null	Web Serv
14	trace	Record	null	Authoriza
15	trace	Record	null	Authoriza
16	trace	Record	null	Authoriza
17	trace	Record	null	Web Servi
18	trace	Record	null	Web Servi
19	trace	Record	null	Web Servi
20	trace	Record	null	Authoriza
21	trace	Record	null	Web Servi
22	trace	Record	null	Authoriza
23	trace	Record	null	Authoriza
24	trace	Record	null	Web Servi
25	trace	Record	null	Authoriza
26	trace	Record	null	Authoriza

Search Columns to Expand

- ☒ (Select All Columns)
- ☒ component
- ☒ alObjectId
- ☒ extensionPublisher
- ☒ sqlRowsRead
- ☒ numberOfDocuments
- ☒ documentFormat
- ☒ alObjectType
- ☒ eventId
- ☒ telemetrySchemaVersion
- ☒ extensionName
- ☒ totalTime
- ☒ sqlExecutes
- ☒ companyName
- ☒ layoutType
- ☒ alStackTrace
- ☒ reportAction
- ☒ Use original column name as prefix

Limit of 1000 scanned rows reached.

OK Cancel

Query Settings

PROPERTIES

Name

Traces

APPLIED STEPS

AnalyticsQuery

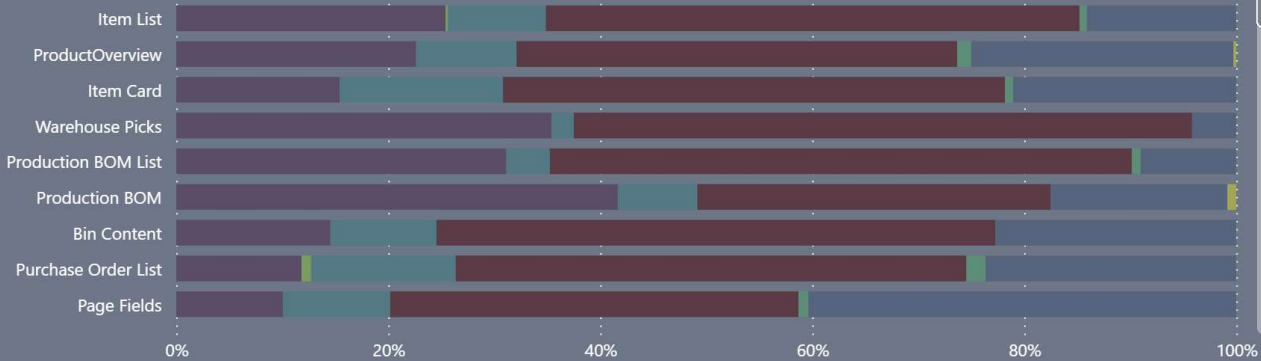
Parsed JSON

32 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

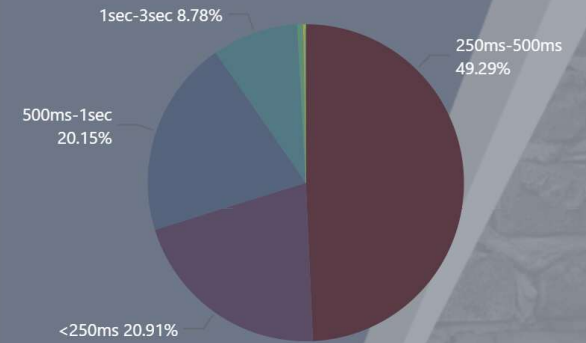


Page Performance

● <250ms
 ● 15sec-30sec
 ● 1sec-3sec
 ● 250ms-500ms
 ● 3sec-7sec
 ● 500ms-1sec
 ● 7sec-15sec

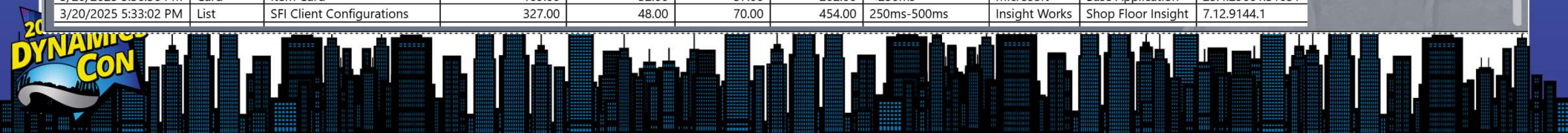


Performance Count



Time Stamp	Page Type	Page	Processing Duration	Receive Duration	Send Duration	Total Duration	Performance Bucket	Publisher	Ext. Name	Ext. Version
3/20/2025 6:58:02 PM	Card	Item Card	234.00	41.00	50.00	336.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:57:57 PM	Document	Purchase Order	218.00	23.00	60.00	313.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:51:35 PM	Card	ProductOverview	195.00	53.00	64.00	317.00	250ms-500ms	null	null	
3/20/2025 6:42:44 PM	List	Production BOM List	360.00	25.00	56.00	446.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:38:41 PM	List	Item List	125.00	44.00	50.00	223.00	<250ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:35:34 PM	Document	Purchase Order	229.00	31.00	60.00	323.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:32:59 PM	List	Item List	151.00	54.00	66.00	278.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:32:11 PM	List	Page Fields	120.00	2.00	36.00	744.00	500ms-1sec	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:28:55 PM	List	Page Inspection	227.00	1.00	95.00	341.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:28:42 PM	Card	Item Card	470.00	13.00	127.00	623.00	500ms-1sec	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:21:59 PM	List	Sales Credit Memos	145.00	26.00	156.00	331.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:20:35 PM	List	Production BOM List	220.00	34.00	74.00	335.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 6:12:10 PM	Card	Item Card	409.00	0.00	77.00	499.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 5:55:45 PM	List	Page Inspection	99.00	49.00	183.00	546.00	500ms-1sec	Microsoft	Base Application	25.4.29661.31031
3/20/2025 5:54:57 PM	Worksheet	Consumption Journal	603.00	30.00	60.00	701.00	500ms-1sec	Microsoft	Base Application	25.4.29661.31031
3/20/2025 5:44:36 PM	List	Purchase Lines	368.00	55.00	78.00	506.00	500ms-1sec	Microsoft	Base Application	25.4.29661.31031
3/20/2025 5:39:29 PM	List	Bin Content	131.00	49.00	78.00	268.00	250ms-500ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 5:36:36 PM	Card	Item Card	109.00	32.00	57.00	202.00	<250ms	Microsoft	Base Application	25.4.29661.31031
3/20/2025 5:33:02 PM	List	SFI Client Configurations	327.00	48.00	70.00	454.00	250ms-500ms	Insight Works	Shop Floor Insight	7.12.9144.1

TM GROUP INC.



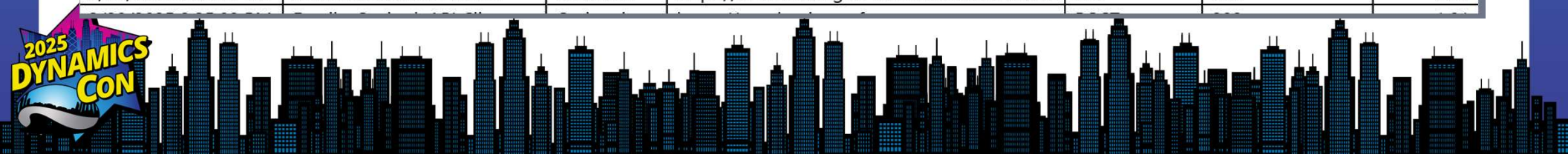
Report Load Performance

TimeStamp EST	Object Name	Reporting Engine	SQL Executes	Rows Read	Server Execution Time	Execution Time	Performance Bucket	Extension Name	Extension Publisher	Extension Version
2/18/2025 2:04:22 PM	WHI Item Barcode Label	Rdlc	1	1	0.50	0.00	250ms-500ms	Warehouse Insight Advanced WMS	Insight Works	2.13.9109.1
2/18/2025 2:05:02 PM	WHI Item Barcode Label	Rdlc	1	1	0.58	0.00	500ms-1sec	Warehouse Insight Advanced WMS	Insight Works	2.13.9109.1
2/18/2025 2:05:17 PM	Calc. Consumption	ProcessingOnly	41	27	0.26	0.00	250ms-500ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:05:28 PM	WHI Item Barcode Label	Rdlc	1	1	0.42	0.00	250ms-500ms	Warehouse Insight Advanced WMS	Insight Works	2.13.9109.1
2/18/2025 2:07:34 PM	Purchase Order	Word	31	46	0.18	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:07:35 PM	Purchase Order	Rdlc	9	29	1.08	0.00	1sec-3sec	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:07:54 PM	Get Source Documents	ProcessingOnly	27	20	0.20	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:08:20 PM	WHI Item Barcode Label	Rdlc	1	1	0.48	0.00	250ms-500ms	Warehouse Insight Advanced WMS	Insight Works	2.13.9109.1
2/18/2025 2:08:23 PM	TMG_CreatePick_BOMRoutLink	ProcessingOnly	191	139	2.80	0.00	1sec-3sec	TMG HME Customizations	The TM Group, Inc.	1.0.0.86
2/18/2025 2:09:06 PM	Whse.-Shipment - Create Pick	ProcessingOnly	36	18	0.77	0.00	500ms-1sec	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:10:39 PM	Whse.-Shipment - Create Pick	ProcessingOnly	45	29	0.18	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:11:05 PM	Whse.-Shipment - Create Pick	ProcessingOnly	24	14	0.10	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:11:24 PM	Whse.-Shipment - Create Pick	ProcessingOnly	23	19	0.04	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:11:53 PM	Whse.-Shipment - Create Pick	ProcessingOnly	23	17	0.05	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:12:22 PM	Get Source Documents	ProcessingOnly	24	19	0.10	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:12:50 PM	Whse.-Shipment - Create Pick	ProcessingOnly	35	27	0.11	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:13:03 PM	TMG_CreatePick_BOMRoutLink	ProcessingOnly	34	23	0.14	0.00	<250ms	TMG HME Customizations	The TM Group, Inc.	1.0.0.86
2/18/2025 2:13:05 PM	Whse.-Shipment - Create Pick	ProcessingOnly	13	11	0.03	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849
2/18/2025 2:13:28 PM	Get Source Documents	ProcessingOnly	41	26	0.16	0.00	<250ms	Base Application	Microsoft	25.2.27733.29849

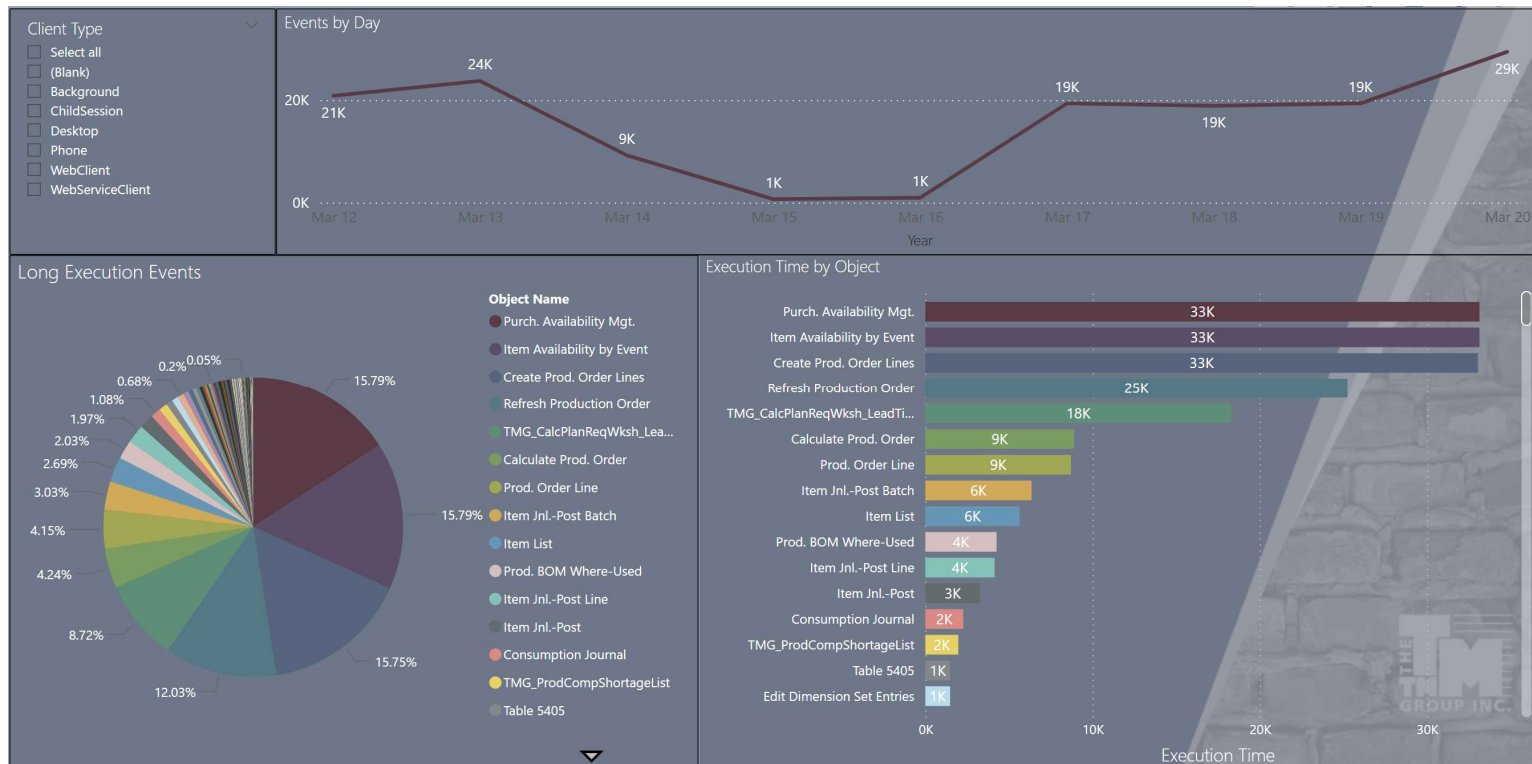


Web Service Performance

Time Stamp EST	Object Name	Object Type	End Point	HTTP Method	HTTP Status Code	Total Time
3/20/2025 2:56:27 PM	IWX PrintNode Management	Codeunit	https://api.printnode.com	POST	201	0.08
3/20/2025 2:56:27 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.07
3/20/2025 2:56:10 PM	IWX PrintNode Management	Codeunit	https://api.printnode.com	POST	201	0.33
3/20/2025 2:56:09 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.13
3/20/2025 2:55:50 PM	Email - Outlook API Client	Codeunit	https://graph.microsoft.com	POST	202	0.23
3/20/2025 2:55:48 PM	Email - Outlook API Client	Codeunit	https://graph.microsoft.com	POST	202	0.19
3/20/2025 2:55:20 PM	Email - Outlook API Client	Codeunit	https://graph.microsoft.com	POST	202	0.94
3/20/2025 2:41:53 PM	IWX PrintNode Management	Codeunit	https://api.printnode.com	POST	201	0.32
3/20/2025 2:41:52 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.03
3/20/2025 2:41:52 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.09
3/20/2025 2:40:28 PM	Email - Outlook API Client	Codeunit	https://graph.microsoft.com	POST	202	0.83
3/20/2025 2:40:28 PM	IWX PrintNode Management	Codeunit	https://api.printnode.com	POST	201	0.08
3/20/2025 2:40:27 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.07
3/20/2025 2:40:13 PM	IWX PrintNode Management	Codeunit	https://api.printnode.com	POST	201	0.24
3/20/2025 2:40:13 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.12
3/20/2025 2:39:57 PM	IWX PrintNode Management	Codeunit	https://api.printnode.com	POST	201	0.08
3/20/2025 2:39:57 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.04
3/20/2025 2:39:41 PM	IWX PrintNode Management	Codeunit	https://api.printnode.com	POST	201	0.08
3/20/2025 2:39:41 PM	IWX Barcode Generation	Codeunit	https://iw-barcodegen-v02.azurewebsites.net	GET	200	0.13



BC Object Performance



SQL Performance

Time Stamp EST	Execution Time	Object Name	SQL Statement
			ASC,"Reservation Status" ASC,"Shipment Date" ASC,"Expected Receipt Date" ASC,"Entry No_" ASC,"Positive" ASC OPTION(OPTIMIZE FOR UNKNOWN, FAST 50)
2/28/2025 3:44:31 PM	0.76	Item List	SELECT "Item"."timestamp" AS "timestamp","Item"."No_" AS "No_","Item"."No_2" AS "No_2","Item"."Description" AS "Description","Item"."Search Description" AS "Search Description","Item" "SQLDATABASE".dbo."CURRENTCOMPANY\$Item\$437dbf0e-84ff-417a-965d-ed2bb9650972" AS "Item" WITH(READUNCOMMITTED) JOIN "SQLDATABASE".dbo."CURRENTCOMPANY\$Item\$4 WITH(READUNCOMMITTED) ON ("Item"."No_" = "Item_ext"."No_") OUTER APPLY (SELECT TOP (1) MAX("TMG_LastDateConsumed\$Item Ledger Entry"."Posting Date") AS "TMG_LastDateCon "SQLDATABASE".dbo."CURRENTCOMPANY\$Item Ledger Entry\$437dbf0e-84ff-417a-965d-ed2bb9650972" AS "TMG_LastDateConsumed\$Item Ledger Entry" WITH(READUNCOMMITTED) WH No_"="Item"."No_" AND "TMG_LastDateConsumed\$Item Ledger Entry"."Entry Type"=@6)) AS "SUB\$TMG_LastDateConsumed" OUTER APPLY (SELECT TOP (1) SUM("Inventory\$Item Ledger E Entry\$SUM\$Quantity" FROM "SQLDATABASE".dbo."CURRENTCOMPANY\$Item Ledger Entry\$437dbf0e-84ff-417a-965d-ed2bb9650972\$VSIFT\$Key2" AS "Inventory\$Item Ledger Entry" WITH(Entry"."Item No_"="Item"."No_")) AS "SUB\$Inventory" OUTER APPLY (SELECT TOP (1) CAST (1 AS TINYINT) AS "Substitutes Exist\$Item Substitution\$EXISTS" FROM "SQLDATABASE".dbo."CURF ed2bb9650972" AS "Substitutes Exist\$Item Substitution" WITH(READUNCOMMITTED) WHERE ("Substitutes Exist\$Item Substitution"."Type"=@7 AND "Substitutes Exist\$Item Substitution"."N TOP (1) CAST (1 AS TINYINT) AS "Stockkeeping Unit Exists\$Stockkeeping Unit\$EXISTS" FROM "SQLDATABASE".dbo."CURRENTCOMPANY\$Stockkeeping Unit\$437dbf0e-84ff-417a-965d-ed2b WITH(READUNCOMMITTED) WHERE ("Stockkeeping Unit Exists\$Stockkeeping Unit"."Item No_"="Item"."No_")) AS "SUB\$Stockkeeping Unit Exists" OUTER APPLY (SELECT TOP (1) CAST (1 AS "SQLDATABASE".dbo."CURRENTCOMPANY\$BOM Component\$437dbf0e-84ff-417a-965d-ed2bb9650972" AS "Assembly BOM\$BOM Component" WITH(READUNCOMMITTED) WHERE ("Asse "SUB\$Assembly BOM" OUTER APPLY (SELECT TOP (1) CAST (1 AS TINYINT) AS "Coupled to Dataverse\$CRM Integration Record\$EXISTS" FROM "SQLDATABASE".dbo."CURRENTCOMPANY\$CF "Coupled to Dataverse\$CRM Integration Record" WITH(READUNCOMMITTED) WHERE ("Coupled to Dataverse\$CRM Integration Record"."Integration ID"="Item"."\$systemId" AND "Coupled "SUB\$Coupled to Dataverse" WHERE ("Item"."No_" COLLATE Latin1_General_100_CI_AI LIKE @0 OR "Item"."Description" COLLATE Latin1_General_100_CI_AI LIKE @1 OR "Item_ext"."TMG_Tr Latin1_General_100_CI_AI LIKE @2 OR "Item"."Description 2" COLLATE Latin1_General_100_CI_AI LIKE @3 OR "Item"."Vendor No_" COLLATE Latin1_General_100_CI_AI LIKE @4 OR "Item"."Ro ("Item"."No_">@9) ORDER BY "No_" ASC OPTION(OPTIMIZE FOR UNKNOWN, FAST 50)
2/28/2025 3:44:30 PM	1.39	Item List	SELECT TOP (50) "Item"."timestamp" AS "timestamp","Item"."No_" AS "No_","Item"."No_2" AS "No_2","Item"."Description" AS "Description","Item"."Search Description" AS "Search Descripti "SQLDATABASE".dbo."CURRENTCOMPANY\$Item\$437dbf0e-84ff-417a-965d-ed2bb9650972" AS "Item" WITH(READUNCOMMITTED) JOIN "SQLDATABASE".dbo."CURRENTCOMPANY\$Item\$4 WITH(READUNCOMMITTED) ON ("Item"."No_" = "Item_ext"."No_") OUTER APPLY (SELECT TOP (1) MAX("TMG_LastDateConsumed\$Item Ledger Entry"."Posting Date") AS "TMG_LastDateCon "SQLDATABASE".dbo."CURRENTCOMPANY\$Item Ledger Entry\$437dbf0e-84ff-417a-965d-ed2bb9650972" AS "TMG_LastDateConsumed\$Item Ledger Entry" WITH(READUNCOMMITTED) WH No_"="Item"."No_" AND "TMG_LastDateConsumed\$Item Ledger Entry"."Entry Type"=@6)) AS "SUB\$TMG_LastDateConsumed" OUTER APPLY (SELECT TOP (1) SUM("Inventory\$Item Ledger E Entry\$SUM\$Quantity" FROM "SQLDATABASE".dbo."CURRENTCOMPANY\$Item Ledger Entry\$437dbf0e-84ff-417a-965d-ed2bb9650972\$VSIFT\$Key2" AS "Inventory\$Item Ledger Entry" WITH(Entry"."Item No_"="Item"."No_")) AS "SUB\$Inventory" OUTER APPLY (SELECT TOP (1) CAST (1 AS TINYINT) AS "Substitutes Exist\$Item Substitution\$EXISTS" FROM "SQLDATABASE".dbo."CURF ed2bb9650972" AS "Substitutes Exist\$Item Substitution" WITH(READUNCOMMITTED) WHERE ("Substitutes Exist\$Item Substitution"."Type"=@7 AND "Substitutes Exist\$Item Substitution"."N TOP (1) CAST (1 AS TINYINT) AS "Stockkeeping Unit Exists\$Stockkeeping Unit\$EXISTS" FROM "SQLDATABASE".dbo."CURRENTCOMPANY\$Stockkeeping Unit\$437dbf0e-84ff-417a-965d-ed2b WITH(READUNCOMMITTED) WHERE ("Stockkeeping Unit Exists\$Stockkeeping Unit"."Item No_"="Item"."No_")) AS "SUB\$Stockkeeping Unit Exists" OUTER APPLY (SELECT TOP (1) CAST (1 AS "SQLDATABASE".dbo."CURRENTCOMPANY\$BOM Component\$437dbf0e-84ff-417a-965d-ed2bb9650972" AS "Assembly BOM\$BOM Component" WITH(READUNCOMMITTED) WHERE ("Asse "SUB\$Assembly BOM" OUTER APPLY (SELECT TOP (1) CAST (1 AS TINYINT) AS "Coupled to Dataverse\$CRM Integration Record\$EXISTS" FROM "SQLDATABASE".dbo."CURRENTCOMPANY\$CF



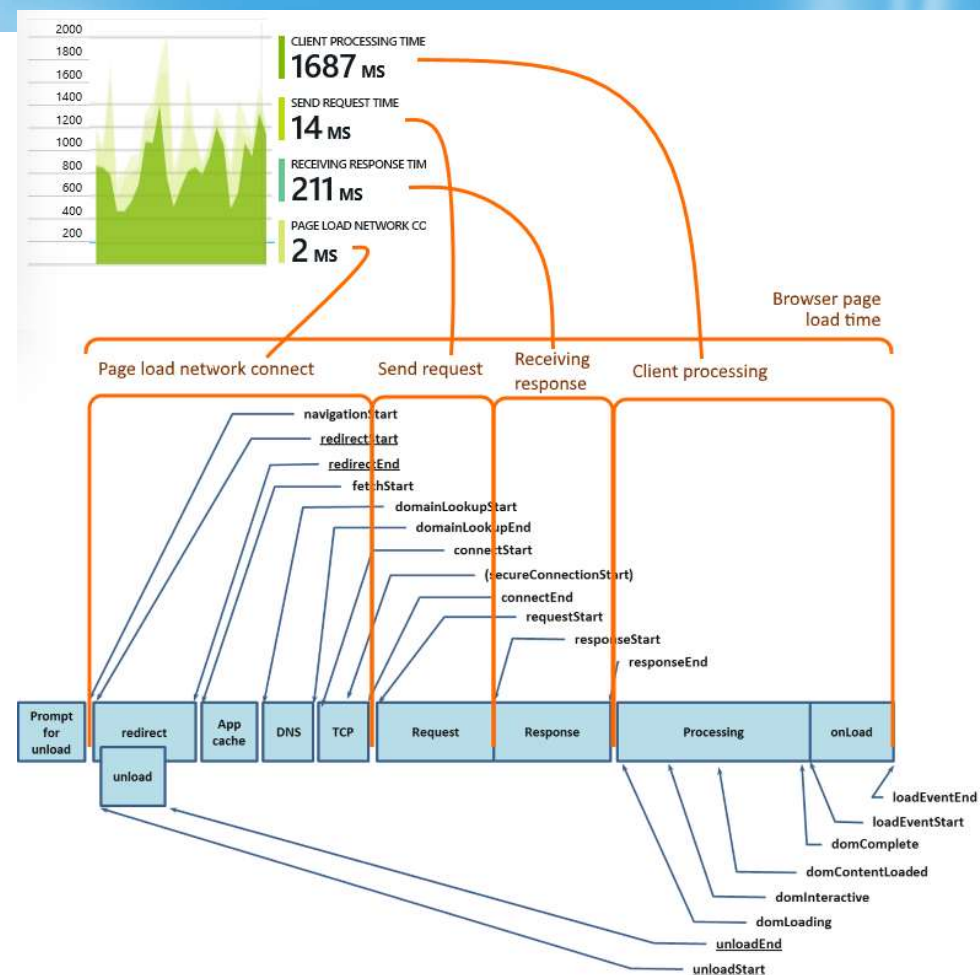
Common Issues

and how to address them.



Network Analysis

- Network Duration
- Send Duration
- Receive Duration
- Processing Duration



Slow Data Retrieval

- Too many rollups
- Too many OnAfterGetRecord calculations
- Insufficient Database Indexing



Database Table Locks

Tell me what you want to do  

database locks

Go to Pages and Tasks

> Database Locks Administration

Seal

 Database Locks





✕ This page shows a snapshot of all database locks. Where possible, it displays details on the AL session that is causing the SQL lock.

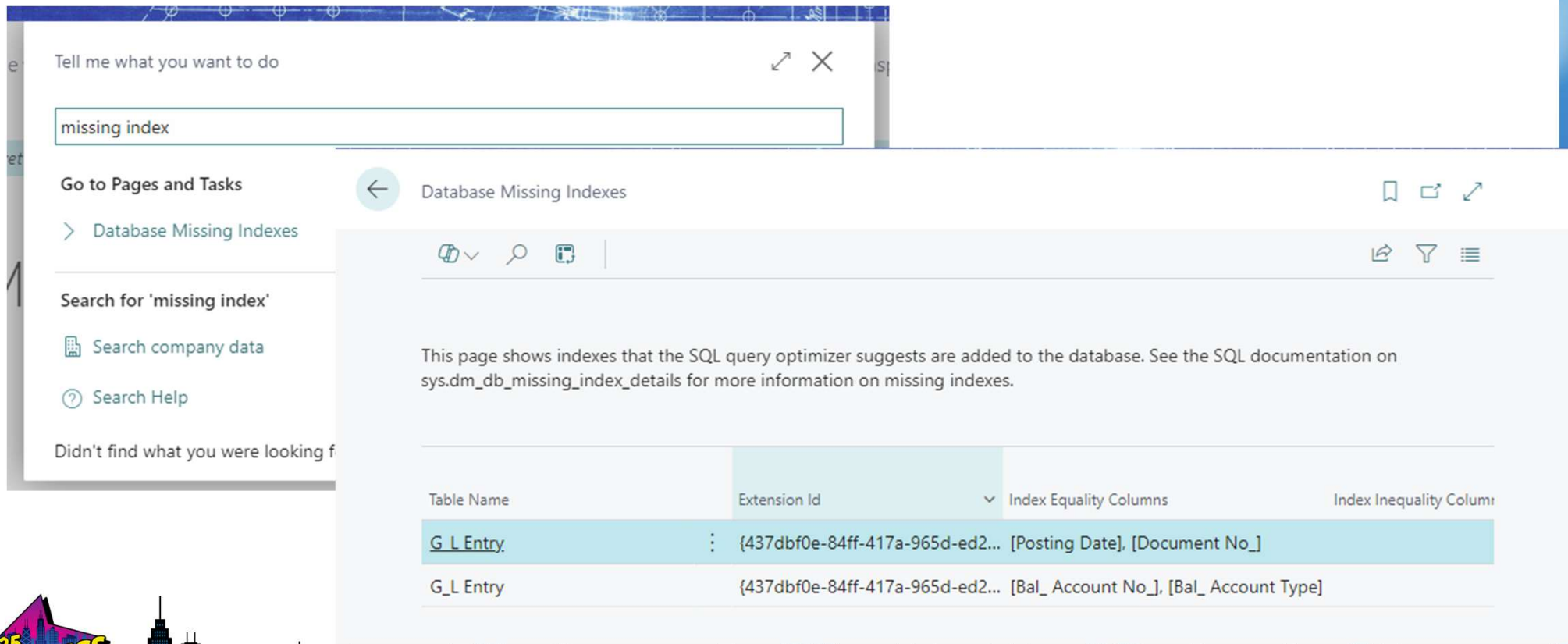




SQL Session ID	Session ID	Table name	SQL Lock Resource Type	SQL Lock Request Mode	SQL Lock Request Status	User Name	Executing AL Object Type	Executing AL Object ID	Executing AL C
(There is nothing to show in this view)									



Database Missing Index



Tell me what you want to do

missing index

Go to Pages and Tasks

> Database Missing Indexes

Search for 'missing index'

Search company data

Search Help

Didn't find what you were looking for

Database Missing Indexes

This page shows indexes that the SQL query optimizer suggests are added to the database. See the SQL documentation on [sys.dm_db_missing_index_details](#) for more information on missing indexes.

Table Name	Extension Id	Index Equality Columns	Index Inequality Columns
G_L Entry	{437dbf0e-84ff-417a-965d-ed2...	[Posting Date], [Document No.]	
G_L Entry	{437dbf0e-84ff-417a-965d-ed2...	[Bal_ Account No.], [Bal_ Account Type]	

Adding Indexes to Business Central

- Added to the Tables in BC
- Needs to be an Extension
- Indexes improve Load Time
- Indexes cost Save Time

```
namespace BCJourney;  
using Microsoft.Finance.GeneralLedger.Ledger;  
  
tableextension 50003 ARD_GLEntry extends "G/L Entry"  
{  
    keys  
    {  
        key(ARD_Index1; "Posting Date", "Document No.")  
        {  
        }  
  
        key(ARD_Index2; "Bal. Account No.", "Bal. Account Type")  
        {  
        }  
    }  
}
```



Flow Field Tax

- Flow Fields perform aggregation over a data set
 - Min, Max, Sum, Lookup...
- Performs a large data query and should be properly indexed
- SIFT Keys are important for Min/Max/Sum calculations
- Can have a high impact when used on a List
- Best on a limited list, card, or fact box



Inefficient OnGetRecord

- Can be used to perform calculation not possible with a Flow Field
- Could be even more costly than a Flow Field
- Runs on Lists in the background, adding data costs
- Use on Cards, sparingly on Lists



Database Locked Tables

- Many processes in BC require exclusive access to a table.
- These should be designed to complete quickly, but...
- Poor indexing could drive these processes to take too long
 - Under indexing OR over indexing
- Moving to scheduled jobs or delegated responsibility



Reporting APIs

- A special designation in the design of an API
- Uses a Read Only version of the tables to avoid locks
- Not an option when exposing a Page as a Web Service



Auto Tuning

- Use the system!
- Microsoft adjusts server resource allocations based on usage
- Don't back off due to slowness during go-live



Questions?

War Stories?



Walkthrough Blog Posts



- Performance Profiler
 - <https://aardvarklabs.blog/2024/10/11/monitor-business-central-performance-with-performance-profiling/>
- VS Code Snapshot
 - <https://aardvarklabs.blog/2024/10/25/development-of-responsive-al-code-with-the-al-performance-tools/>
- Application Insights
 - <https://aardvarklabs.blog/2024/10/04/monitor-business-central-performance-with-application-insights/>



Thank you

Find me at AardvarkLabs.blog

The logo for '2025 DYNAMICS CON' is displayed in a bold, yellow, stylized font with black outlines. The text is curved to follow the shape of a large, grey, stylized letter 'D' that serves as a background. Behind the 'D' is a blue city skyline with various building silhouettes. The entire graphic is set against a solid blue background.

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